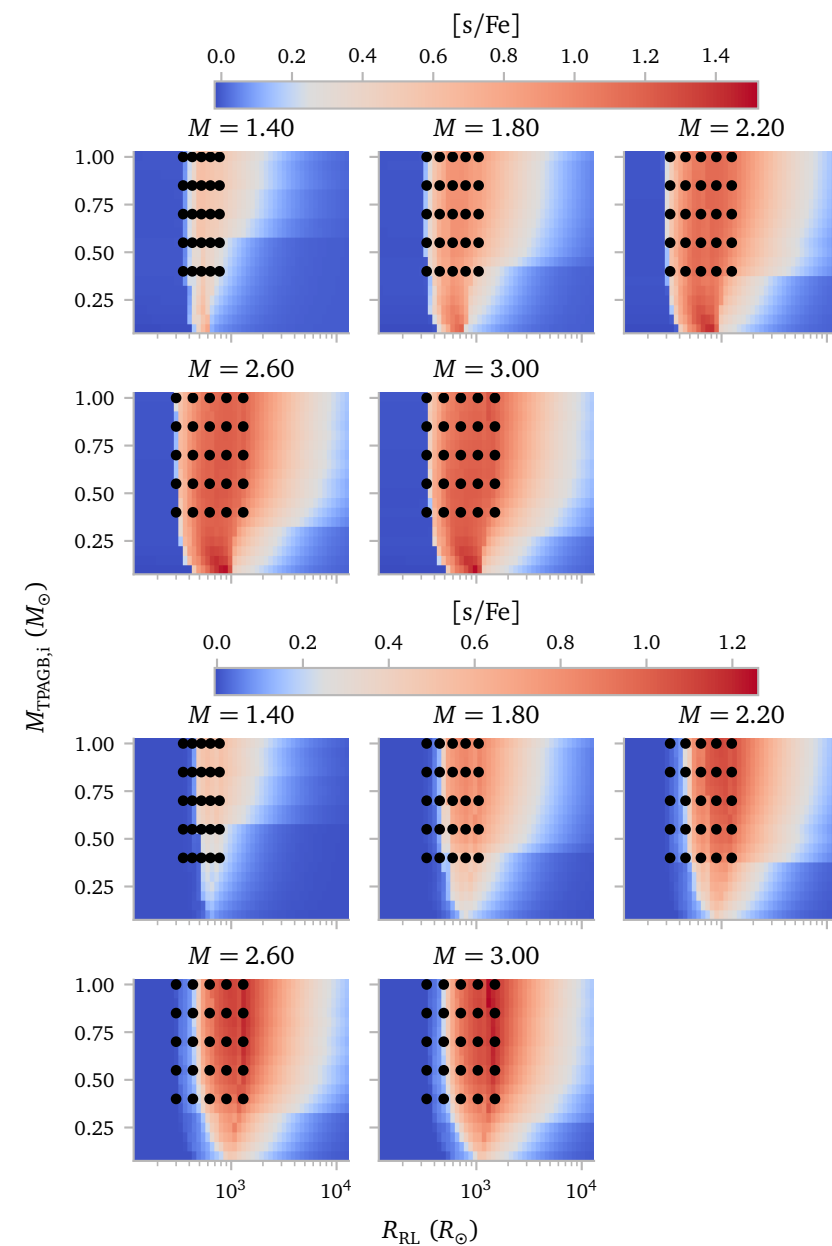


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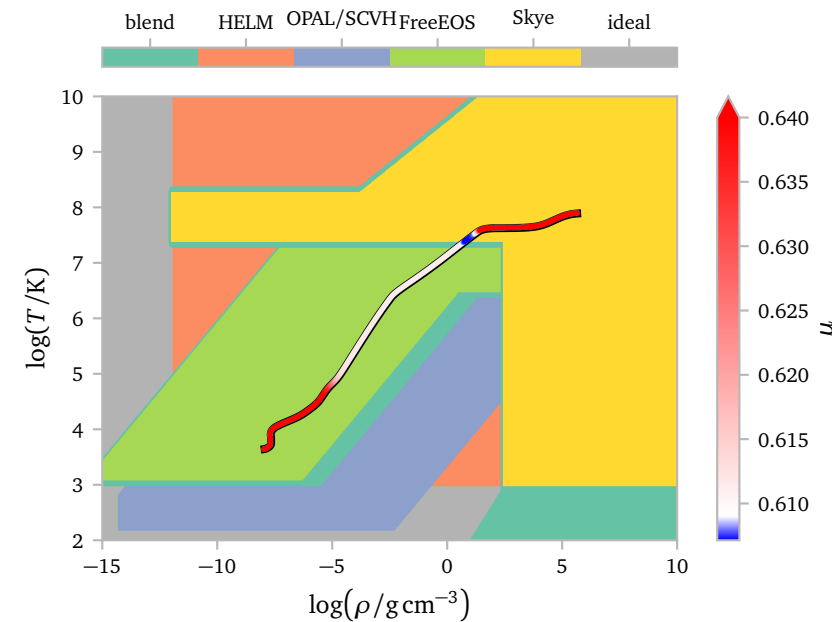
a. Grid



b. Mean Molecular Weight

b.1 Giant Branch Dip

Using the `EOS_plotter` routine from *MESA*, it is possible to clearly plot the different EOS tables that *MESA* uses, and in which regions in the ρ - T diagram. Below, I show this, and I plot the ρ - T profile for an evolved RGB star over these tables.



The RGB profile is colored, where very red regions correspond to regions with relatively large mean molecular weight μ . Blue regions correspond to the lowest mean molecular weights, and white corresponds to an average mean molecular weight of the envelope. I think this figure quite clearly shows why we see a pit in the mean molecular weight. The core boundary happens inside the Skye EoS table (Jermyn et al., 2021). This table *assumes* complete ionization. Once the profile crosses over to the boundary region, we see that the mean molecular weight increases until it crosses the *FreeEOS* regime, which does *not assume* complete ionization.

References

Jermyn, A. S., Schwab, J., Bauer, E., Timmes, F. X., & Potekhin, A. Y. (2021) *Skye: A Differentiable Equation of State*. *ApJ***913**, 72.